Unsupervised Wavetable Seam Artifact Repair via Hybrid GA–PPO Meta-Search

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Code (ReelSynth engine + DenoiseOpt meta-search): https://github.com/reeldemo/reelsynth, https://github.com/reeldemo/denoise-opt-meta

ABSTRACT

In single-period wavetable synthesis, a sample discontinuity at the wrap boundary injects a broadband transient that repeats at the fundamental frequency, producing an audible periodic click. We address this *cycle-local seam artifact* with **DenoiseOpt**, a residual-scored hybrid meta-search that selects and fits bake-cell operators applied at the period boundary. The bake cell composes classical DualCosine crossfades, FIR blends, and optional lightweight residual networks, possibly routed via Mixture-of-Experts (MoE) soft gating, without requiring paired studio-clean recordings. Candidates are ranked by the *prolonged residual score* $R \in [0,1]$ (higher is better): the clamped tiled-RMS agreement between each method's output and a procedural ideal sibling $r^\star$ (open-wrap cliff withheld). No-bake leaves the cracked engine unchanged and scores that unrepaired period against the same $r^\star$; it is not the ideal itself. The outer search loop interleaves genetic algorithm (GA) tournament selection, PPO-based architecture proposals, and optional population-based training (PBT). On a frozen canonical sine+cliff holdout (seed `20260719`), the compact meta-favorite achieves $R \approx 0.991$ vs classical no-bake $R \approx 0.972$, `seam_fir3` $R \approx 0.961$, and DualCosine $R \approx 0.825$ (gaps vs classical board; objective is maximize R toward the ideal sibling). Hard-cliff strata (top 10% wrap-jump, n=410) keep favorite $R \approx 0.9915$ while DualCosine falls to $R \approx 0.657$; no-bake R stays high (≈ 0.967) because residual mass is small relative to ideal RMS (unrepaired engine / passthrough, $x \mapsto x$). A same-generator corrupt→corrupt Noise2Noise control reaches $R \approx 0.992$; a supervised LSTM sibling ceiling reaches $R \approx 0.995$. Wavetable-native transfer uses true ReelSynth-exported factory periods (n=25) and external AKWF CC0 single-cycle WAVs (n=24) under the same wrap protocol (no LibriSpeech/MUSDB). On exports the favorite trails no-bake / FIR (domain shift); on AKWF it matches no-bake and beats DualCosine. Scope is strictly cycle-local wavetable seam restoration aimed at DSP / synthesis venues (DAFx / AES / arXiv cs.SD).

Keywords unsupervised learning; wavetable; wrap discontinuity; seam restoration; DenoiseOpt; prolonged residual score; neural architecture search; reinforcement learning; genetic algorithms; meta-learning

1 Introduction

Wavetable oscillators store a single period of a waveform and advance a read pointer modulo the table length. When $x[0] \neq x[L-1]$, the wrap is a hard discontinuity that, under cyclic playback, becomes a periodic click with a broadband spectrum locked to the fundamental. The same geometry appears whenever a short loop buffer is tiled for listening or for objective evaluation. We define the repair task as *cycle-local seam restoration*: edit a neighborhood of the wrap (and optionally a lightweight residual cell over the period) so that multi-period playback matches an ideal reference that never opens the wrap cliff. That outer agreement is scored by the prolonged residual R for this periodic seam artifact class.

Figure 1 shows one real sine-wrap edge case from the frozen canonical holdout (same `make_batch` tile, seed `20260719`). Panel (a) is the ideal multi-period sine. Panel (b) is the cracked engine tile after open-wrap cliff modulation ($R \approx 0.950$), with the ideal overlaid so the seam jump is visible. Panel (c) applies DualCosine on that same tile ($R \approx 0.603$ on this severe cliff). Panel (d) is the top DenoiseOpt favorite (`champion_iter_000235`, batch $R \approx 0.991$) on the same tile ($R \approx 0.9917$). Panels (e)–(f) plot the modulation residual and a seam zoom so the favorite's correction is readable against DualCosine.

Classical virtual-analog practice already supplies seam-local tools. Alias-free BLIT/BLEP-style synthesis [1], PolyBLEP / fractional-delay antialiasing [3], and BLAMP corner corrections [2] reduce wrap and geometric discontinuity energy in-band. Those operators act locally; we score cycle-local bake approximations of them beside DualCosine (Section 5.7). They leave open which bake parameters and which residual cell, if any, minimize prolonged tiled error against a no-cliff ideal. Label-free restoration work such as Noise2Noise shows that useful reconstructors can be ranked

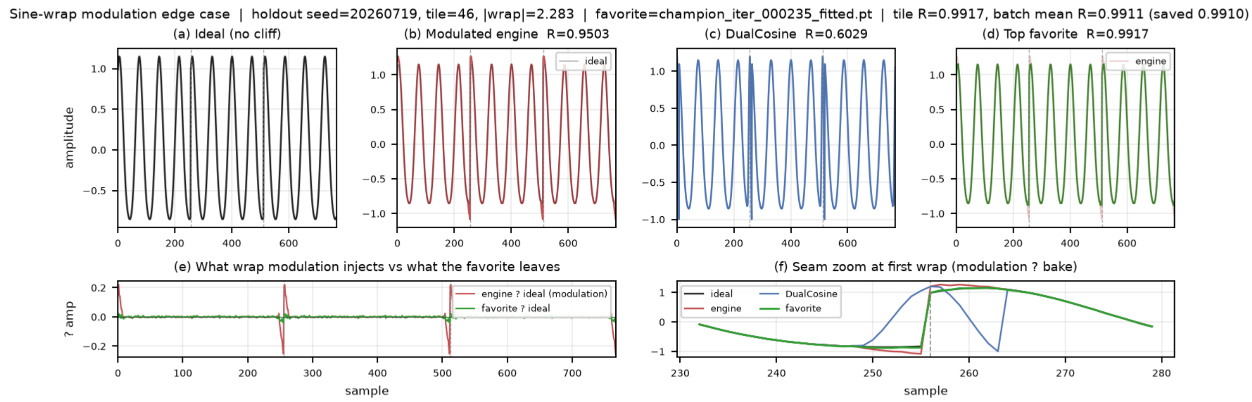

Figure 1: Canonical holdout sine-wrap edge case (seed `20260719`, L=256, N=16 tiles, same tile across panels). (a) Ideal sibling (cliff withheld). (b) Engine with open-wrap cliff (ideal overlay). (c) DualCosine bake. (d) DenoiseOpt favorite bake. (e)–(f) Modulation residual and seam zoom. Prolonged residual $R \in [0, 1]$ (1 = best) is printed per panel where scored. Accessibility: six-panel waveform comparison of ideal, cracked, DualCosine, and favorite seam repairs.

without paired clean targets when the evaluation geometry is explicit [4]. Speech-domain Noise2Noise denoisers extend that idea to audio waveforms [5]. Mixture-invariant unsupervised separation likewise avoids paired clean stems for ranking [29]. On the search side, NAS surveys [16], RL controllers [17, 18], aging evolution [20, 21], population-based hyperparameter schedules [22], CMA-ES tutorials [24], and evolution-guided policy gradients [23] motivate a hybrid outer loop: GA for population diversity, PPO for discrete architecture edits [19], and PBT-style exploit–mutate on elites. Platform-aware and progressive NAS add compute-aware search priors [27, 28]. MoE soft gates suggest routing among heterogeneous tiny cells under one residual score [25]. Waveform separation nets (Wave-U-Net, Conv-TasNet, dual-path RNN, SepFormer) and audio DL surveys inform the searchable cell family at bake-scale width [7–9, 11, 12, 30, 31], with U-Net / ResNet / attention primitives as block ancestors [10, 13–15].

The contribution is a meta-learning protocol for this periodic seam instance. An outer RL+GA(+PBT) loop proposes bake-cell operator graphs and continuous bake parameters. Each candidate is scored by prolonged residual $R \in [0, 1]$ (1 = best) between engine-baked multi-period audio and a procedural ideal tiling. Paired studio-clean wavetables are not required for every trial. Differentiable DSP modules [6] place DualCosine / FIR / MLP seam ops in an interpretable stack. Bayesian HPO [26] informs prior families that compete under the same R. A CUDA hybrid search campaign (PPO+GA+PBT+NAS, depth and MoE biases) has passed a 5,000-iteration clean gate with champion $R \approx 0.9909$ versus DualCosine ≈ 0.8166 on the runner geometry. The frozen canonical holdout and inference benches in Section 5 are the measured evaluation matrices for this report. We do not claim general speech or music enhancement SOTA.

1.1 Contributions

1. **Problem framing.** Cast wavetable wrap discontinuity as cycle-local seam restoration under prolonged tiled evaluation, separating seam-local diagnostics from residual R.
2. **Hybrid DenoiseOpt outer loop.** Specify RL+GA(+PBT) search over discrete operator graphs and continuous bake parameters, with named branches (GA crossover, PPO mutation, PBT exploit, MoE soft gates).
3. **Formal residual protocol.** Define ideal-tile siblings, prolonged residual R, and reimplementable Algorithms 1–9. No wrap-closure or search-convergence guarantees.
4. **Open evaluation path.** Release companion code for the synthesizer engine and meta search, with frozen 5k-gate figures, hard-cliff strata, N2N/seq baselines, and wavetable-native transfer benches.

Section 2 organizes related work by mechanism. Sections 3–4 detail the residual objective and search protocol. Section 5 reports the 5k-gate freeze and inference benches.

2 Related Work

We organize prior work by mechanism. **Used** lines are methodological or comparative anchors for residual-scored RL+GA meta-search. **Screened** lines were inspected as contrast. They are excluded from the dependency set. Paraphrases prefer attached abstracts and PDF snippets over title-only citation.

2.1 Discontinuity control and modular audio DSP

Alias-free digital synthesis of classic analog waveforms (BLIT/BLEP lineage) states the wrap discontinuity problem for trivial oscillators [1]. PolyBLEP-style fractional-delay corrections give practical antialiasing oscillators for subtractive synthesis [3]. BLAMP corner rounding treats discontinuities in the first derivative (triangular edges, clipping, rectification) without heavy oversampling [2]. We score and illustrate cycle-local approximations of those three operators beside DualCosine on the same prolonged-R protocol (Figure 5, Section 5.7). They are no longer citation-only anchors. DDSP reframes DSP modules as composable blocks inside learning pipelines [6]. We use that framing for small bake/FIR/MLP seam cells. We do not train end-to-end neural vocoders for every meta trial. These works are **used** as domain anchors for the operator vocabulary, as scored classical controls, and for why prolonged cyclic playback is the outer judge.

2.2 Label-free restoration

Noise2Noise shows that restoration mappings can be learned from corrupted observations alone by exploiting structure in the corruption process [4]. Speech Noise2Noise denoisers apply the same idea to waveforms without clean paired audio [5]. Mixture invariant training ranks unsupervised separation systems without stem labels [29]. That philosophy motivates ranking denoise candidates without paired studio-clean wavetable cycles. Earlier drafts of this work cited Noise2Noise only as ranking philosophy and omitted a same-domain learned control. We now train a true corrupt→corrupt N2N-style seam restorer on the same L=256 `make_batch` geometry (primary baseline), plus a sibling-supervised ceiling and lightweight LSTM / 1D-CNN seq models (Section 5). Prior speech N2N restorers operate on broadband noisy speech. We operate on periodic seam cliffs with procedural siblings. We **use** this family as conceptual justification for unsupervised residual ranking and as an explicit learned contrast to DenoiseOpt meta-search. Full Demucs / DiffWave stacks are screened: domain mismatch (full-band speech/music denoisers vs short periodic seams), compute, and OA/eval protocol mismatch. We do not retrain image Noise2Noise CNNs.

2.3 Waveform denoising and separation cells

Deep learning for audio surveys the move from hand-designed features to learned time/frequency models [7]. Wave-U-Net performs end-to-end source separation with multi-scale 1-D U-Nets [8]. Improved Wave-U-Net speech enhancement and singing-voice U-Nets popularized encoder–decoder skips on audio [9, 10]. The original U-Net, ResNet residuals, and Transformer attention supply the block primitives we shrink to bake width [13–15]. Conv-TasNet and dual-path RNN demonstrated strong time-domain separation with temporal convolutions and long-context paths [11, 12]. SepFormer and audio spectrogram transformers push attention-heavy audio stacks further [30, 31]. These lines are **used** as design priors for tiny searchable cells (shallow U-Net / dilated / dual-path / attention blocks under a strict per-trial bake budget). They are not frozen full-scale separators inside every meta trial.

2.4 NAS and RL controllers

Elsken et al. taxonomize NAS by search space, strategy, and evaluation cost [16]. Their performance-estimation axis includes lower-fidelity proxies such as shorter training, data subsets, and lower-resolution images [16]. Zoph and Le couple RL controllers to architecture search [17]. ENAS adds parameter sharing for efficient discrete search [18]. Progressive and platform-aware NAS add staged evaluation and latency-aware rewards [27, 28]. We **use** this family as the ancestor of RL proposing architecture edits (add/remove/mutate ops, toggle residual-primary, mutate MLP/FIR), scored with prolonged residual. PPO supplies the clipped-surrogate policy update when the hybrid search loop logs PPO mutation branches [19].

2.5 Evolutionary NAS and population schedules

Large-scale and regularized / aging evolution showed that evolutionary NAS can discover competitive image classifiers when population turnover is carefully regularized [20, 21]. Population Based Training jointly adapts weights and hyperparameters in an asynchronous population [22]. Practical Bayesian optimization [26] and CMA-ES tutorials [24] inform local Bayesian and continuous evolutionary prior families. These are **used** for GA tournament/crossover branches, PBT-style exploit–mutate, and literature-combo priors, implemented at synthesizer-trial scale.

2.6 Evolutionary reinforcement learning hybrids

Khadka and Tumer's evolution-guided policy gradient (ERL) interleaves an evolutionary population with an RL actor to mitigate sparse reward and brittle hyperparameters in deep RL [23]. We **use** ERL as the spirit of interleaving GA population steps with PPO updates at tiny population sizes. We do not claim a full ERL robotics stack.

2.7 Mixture-of-experts soft gating

Sparsely gated MoE layers increase capacity via conditional computation [25]. We **use** MoE as motivation for soft gates over heterogeneous seam blocks (e.g., U-Net vs dilated vs attention cells) under residual scoring, at bake-scale widths.

2.8 Used versus screened

Used (methodological or comparative anchors).

- Discontinuity / VA lineage: [1–3].
- Unsupervised restoration philosophy: [4, 5, 29].
- Modular / differentiable audio DSP context: [6].
- Audio DL and compact waveform cells: [7–15, 30, 31].
- NAS / RL controllers / PPO: [16–19, 27, 28].
- Evolutionary NAS and population HPO: [20–22, 24, 26].
- ERL hybrid outer loop: [23].
- MoE soft gating prior: [25].

Screened (contrast only).

- **MAML** [32]: bi-level / fast-adaptation framing. We do not differentiate through a deep task network. Inner fits are coordinate-style updates on bake parameters. Outer rank is residual R.
- **DARTS** [33]: continuous relaxation of architecture search. We keep discrete ops + small cells under residual scoring.
- **Demucs / Hybrid Demucs / realtime waveform enhancement** [34–36]: strong separators and enhancers, but full multi-scale / hybrid STFT stacks are too heavy per meta trial under bake-cell search budgets.
- **DiffWave** [37]: diffusion vocoder. Screened as a generative sampling stack outside the seam bake operator set.
- **SEGAN-style adversarial enhancement** [38]: optional tiny adversarial cues only. Full GAN training loops screened for per-trial cost.

Positioning. Relative to this map, the present work contributes (i) a prolonged residual objective for periodic seam denoising, (ii) an explicitly hybrid RL+GA(+PBT) meta-outer loop with honest branch naming, and (iii) a used/screened literature protocol with downloaded source PDFs. Claims stay inside cycle-local seam restoration under a declared compute budget. The primary claim is protocol-level: residual-scored meta-search for cycle-local audio denoise operators.

3 Methods

3.1 Notation

Throughout, $x \in \mathbb{R}^L$ is the stored single-period wavetable (raw engine input). $y = \Theta(x;\theta)$ is the engine-baked cycle after seam ops and optional residual cells with parameters θ. $r^\star \in \mathbb{R}^L$ is a procedural ideal sibling from the same seed and family with the open-wrap cliff withheld. Studio-clean recordings are not used as $r^\star$. Tiling length N (typically N=16) forms $y_{\text{tiled}} = \text{Tile}(y,N)$ and $r^\star_{\text{tiled}} = \text{Tile}(r^\star,N)$. Seam width W (code: `SEAM_W`, typically 8) is the neighborhood edited by classical fade and polish ops. $R \in [0,1]$ is the prolonged residual score defined below (1 = best). These symbols are fixed for the residual objective. They follow a different convention from Noise2Noise clean/noisy pairs.

3.2 Ideal tile construction

The ideal sibling is generated by the same procedural draw as the engine input (frequency, phase, mid-cycle noise policy) with the open-wrap cliff amplitude set to zero at both ends. Engine and ideal therefore share the mid-cycle content of one seed draw. They differ by the seam cliff (and any seam-boosted noise policy applied only to the engine path). This sibling relationship is what makes unsupervised ranking possible without paired studio recordings [4, 5].

Table 1: Core symbols used in Methods.

Symbol	Meaning
L	Cycle length (samples per period)
x	Raw engine wavetable period
$y = \Theta(x;\theta)$	Baked cycle after seam ops / residual cell
$r^\star$	Ideal sibling (open-wrap cliff withheld)
N	Prolong tiling count
W	Seam neighborhood width
R	Prolonged residual score in $[0,1]$ (1 = best)
θ	Continuous bake / residual parameters

Algorithm 1 ResidualScore

Require: Ideal cycle $r^\star$, baked cycle y, periods N

1: $I \leftarrow \text{Tile}(r^\star, N)$
2: $Y \leftarrow \text{Tile}(y, N)$
3: $r_{\text{rms}} \leftarrow \text{RMS}(Y - I)$
4: $i_{\text{rms}} \leftarrow \max(\text{RMS}(I), \varepsilon)$
5: **return** $\text{clamp}(1 - r_{\text{rms}}/i_{\text{rms}}, 0, 1)$

3.3 Residual score R

Every method—no-bake, DualCosine, FIR, poly, VA residuals, neural favorite—is scored the same way: apply an operator Θ to the cracked engine period x to obtain $y = \Theta(x)$, then compare the tiled y to the *same* ideal sibling $r^\star$ under (1). The ideal sibling is the scoring target (open-wrap cliff withheld). It is *not* a method output and it is *not* the no-bake baseline.

With tiling length N,

$$R = \text{clamp}\left(1 - \frac{\text{rms}(y_{\text{tiled}} - r^\star_{\text{tiled}})}{\max(\text{rms}(r^\star_{\text{tiled}}), \varepsilon)}, 0, 1\right), \tag{1}$$

where $\text{rms}(z) = \sqrt{\frac{1}{|z|}\sum_i z_i^2}$ and $\varepsilon > 0$ avoids division by zero. By construction $R \in [0,1]$, R=1 when $y_{\text{tiled}} = r^\star_{\text{tiled}}$, and (when pre-clamp scores lie in $(0,1)$) R is strictly decreasing in residual RMS for fixed ideal RMS. A method "beats" another iff its R is higher on the same $(x, r^\star)$ draw—equivalently, iff its bake is closer to the ideal sibling. Optional soft shape gates (when enabled) multiply R by factors that down-weight mid-cycle damage relative to seam-local error. We do not claim convergence guarantees for the hybrid GA+PPO+PBT outer loop, nor a general wrap-closure theorem for arbitrary bake cells. Empirical gains are reported under the frozen protocol. Outer-loop cost scales as population size × proposals per iteration × fit steps × batch forward cost (order 10^3–10^4 arch evaluations at the reported 5k gate). We report measured budgets, not a convexity claim.

Remark 1 (Noise2Noise motivation). Ranking without studio-clean pairs is possible when the ideal is a procedural sibling of the engine draw [4, 5]. The remark is a design philosophy for procedural siblings.

3.4 Bake operator and baselines

A **bake cell** $\Theta(\cdot;\theta)$ composes classical seam ops (DualCosine, polish, soft seam, FIR blends) with optional tiny residual cells

Algorithm 2 DualCosine baseline

Require: Engine cycle x

Fade both ends of x with raised-cosine windows of width W

return ResidualScore$(r^\star, x_{\text{faded}}, N)$

Algorithm 3 No-bake (passthrough) control

Require: Engine cycle x (open-wrap cliff present)

return ResidualScore$(r^\star, x, N)$ ▷ no operator: $x \mapsto x$

Algorithm 4 MoE soft gating over bake experts

Require: Cycle x, expert cells $\{E_k\}_{k=1}^{K}$, gate logits $g \in \mathbb{R}^K$

$w \leftarrow \text{softmax}(g)$

$y \leftarrow \sum_{k=1}^{K} w_k E_k(x)$

return y

(MLP/FIR/U-Net-lite/attention-lite) and optional MoE soft gates over experts. Bake parameters θ live in a bounded box. Architecture strings are discrete graphs over those ops. Separate classical controls apply cycle-local BLIT/BLEP, PolyBLEP, and BLAMP residual corrections at the wrap on the cracked engine tile (same interface as DualCosine / FIR; not a full oscillator rewrite). Plain-language roles: BLIT/BLEP and PolyBLEP correct a *value* step at wrap. BLAMP corrects a *slope*/corner jump. DualCosine fades endpoints without a BLEP residual. They are scored and illustrated in Section 5.7.

No-bake (passthrough) control. The **no-bake** baseline is the unrepaired cracked engine: $\Theta_{\text{no-bake}}(x) = x$, then $R =$ ResidualScore$(r^\star, x, N)$. It is the engine period *as stored*, scored against the ideal sibling. It is *not* the ideal sibling itself, not a residual-network skip, not a learned identity map, and not a repair method. On mild cliffs residual mass $\ll$ ideal RMS under (1), so no-bake R can sit near 0.97 while the wrap click remains audible—a reference floor/ceiling for "do nothing," not a perceptual wrap score. Released JSON artifacts may still use the legacy key `identity` for this control; manuscript tables and prose use *no-bake*.

Classical (non-AI) comparison board. Learned / searched methods are always ranked against the full classical board on absolute R (same $r^\star$): no-bake, DualCosine / cosine fade, `seam_fir3`, classic quadratic, hann/crossfade/soft fades, detrend/ensemble, poly seam, and cycle-local BLIT/BLEP / PolyBLEP / BLAMP where scored. The optimization objective is to *maximize* prolonged R toward the ideal sibling (best attainable R=1 under (1)); higher R is strictly better. DualCosine appears in PPO as a *constant centering* term ($\rho = R - R_{\text{DualCosine}}$) so advantages are roughly zero-mean early in search—it is not the reward target and not the sole classical comparator. When tables show ΔR vs DualCosine, that is a reporting gap vs one classical bake, not the definition of reward.

Algorithm 5 FitCell (inner loop)

Require: Architecture cfg, fit steps T, batch size B

cell $\leftarrow$ SeamCell(cfg); opt $\leftarrow$ Adam(cell.params)
prev $\leftarrow$ null; patience $\leftarrow 0$
for $t = 1, \ldots, T$ **do**
  ideal, eng $\leftarrow$ MakeBatch(B)
  out $\leftarrow$ ApplyOps(eng, cell, cfg.ops)
  $R \leftarrow$ mean ResidualScore(ideal, out)
  loss $\leftarrow 1 - R$
  if finite(loss) **then** AdamStep(opt, loss)
  end if
  if prev $\neq$ null and $|\text{prev} - R| / \max(|\text{prev}|, \varepsilon) < 10^{-4}$ **then**
    patience $\leftarrow$ patience $+1$; **if** patience ≥ 3 **then break**
  else
    patience $\leftarrow 0$
  end if
  prev $\leftarrow R$
end for
return cell, R

3.5 Search space

Each genome encodes: ordered op list (fade, polish, FIR, pin, residual write), depth (number of composed blocks), residual cell type (none, MLP, FIR, U-Net-lite, attn-lite, LSTM-lite), optional MoE expert count $K \in \{2, 3, 4\}$, and continuous θ in box bounds used by the CUDA search runner (fade widths, FIR taps, residual scales). The searchable bake vocabulary includes a width-capped 1–2 layer LSTM block over the seam window or full cycle (distinct from the fixed supervised seq-LSTM SOTA baseline). The live hybrid tag is `PPO+GA+PBT+NAS+depth+MoE` with LSTM enabled in the block graph.

3.6 Training / fit inner loop

3.7 Hybrid outer loop

Each iteration rotates among branches: PPO, GA, PBT, NAS, and combo. Selection and FitCell are always by absolute residual R (maximize toward the ideal sibling; loss $1 - R$). The reward is larger when R is closer to the best possible value ($R \to 1$).

Near-ceiling R and why reward / search HPs matter. On the sine+cliff generator, no-bake already sits near $R \approx 0.97$, and strong classical / searched bakes live in a narrow band toward 1 (often 0.99–0.991). Absolute gaps of order 10^{-3}–10^{-2} are scientifically real (closer to $r^\star$) but tiny as raw PPO/GA signals: without shaping, advantages collapse and hyperparameters (clip, entropy, mutation rate, population size, fit steps, learning rate) dominate whether the outer loop can climb the last percent. We therefore treat **reward shaping and search hyperparameters as first-class tunables** alongside bake architecture—not as fixed afterthoughts. In the present hybrid, PPO uses a searchable shaping mode among

Algorithm 6 GA tournament step

Require: Population P, tournament size k, crossover rate p_c, mutation rate p_m
```
Sample k genomes; parents ← top-2 by R
child ← Crossover(parents) with prob. p_c, else clone elite
Mutate(child) with rate p_m (ops, depth, cell type, θ)
Fit and evaluate child; insert into P (replace worst if full)
return updated P
```

Algorithm 7 PPO architecture update

Require: Actor–critic π_ϕ, clip ε, experience buffer of mutate actions
```
Sample action a editing ops / hyperparameters from π_φ
Fit proposed cfg; observe centered advantage ρ = R − R_DualCosine    ▷ objective remains maximize R
Advantage Â from critic; ratio r_φ = π_φ(a)/π_φold(a)
Update φ with clipped surrogate min(r_φ Â, clip(r_φ, 1 − ε, 1 + ε)Â)
return updated π_φ
```

Algorithm 8 PBT exploit–mutate

Require: Population P, exploit quantile q, mutate scale σ
```
Rank P by R; for each bottom member, copy hyperparameters from a top-q elite
Perturb copied hyperparameters by multiplicative noise of scale σ
Refit briefly; write back into P
return updated P
```

Algorithm 9 HybridMetaSearch

Require: Iteration budget T_{out}, population size n
```
P ← InitPopulation(n); π ← ActorCritic(); champ ← null; boredom ← 0
for it = 1, ..., T_out do
    branch ← Rotate(ppo, ga, pbt, nas, combo)
    cfg, hp ← Propose(branch, P, π)
    cell, R_fit ← FitCell(cfg, hp.fit_steps)
    R_eval ← EvalCell(cell)
    R ← ½ R_fit + ½ R_eval (+ optional depth/MoE bonus)
    UpdatePopulation(P, cfg, R); PPOUpdate(π, centered ρ = R − R_DualCosine)
    if R > champ.R then champ ← (cfg, cell, R); boredom ← 0
    else boredom ← boredom + 1
    end if
    if boredom ≥ 1000 then PlateauAdapt; boredom ← 0
    end if
end for
return champ
```

{`abs_r`, `vs_dualcosine`, `vs_nobake`, `neglog_gap`} (PBT-mutated; default $\rho = R - R_{\text{DualCosine}}$). DualCosine centering is one shaping choice, not the objective. PBT mutates fit / PPO hyperparameters and reward mode in-population; plateau adaptation retunes branch mix and depth/MoE bias when champions stall. The matched 5k meta-approach comparison (Section 3.8) is the sole GPU experimentation vehicle for outer-loop family, HP, and reward-shaping sweeps under a shared FitCell budget (no concurrent long hybrid campaign).

3.8 Meta-learning approach comparison

Beyond the hybrid outer loop, we compare five additional outer-loop families under a matched 5,000-evaluation budget, shared search seed `1902771841`, and the same FitCell / prolonged-R protocol (LSTM and xLSTM included in the searchable bake vocabulary). Table and figure legends use the short names below (artifact folders keep stable code keys; see `NOMENCLATURE.md`):

- **Random NAS** (`random`): independent uniform draws over bake graphs and fit hyperparameters (control).
- **Cont. CMA-ES** (`cmaes`): diagonal covariance-matrix adaptation over a continuous encoding of depth, width, cell kind, block logits, and fit hyperparameters [24].
- **Arch REINFORCE** (`reinforce`): vanilla policy gradient over discrete architecture-edit actions (no clipped surrogate; contrast to PPO [17, 19]).
- **Aging evolution** (`aging_evo`): regularized evolution with oldest-member replacement after tournament parent selection [20, 21], distinct from the hybrid's non-aging tournament GA branch.
- **TPE Bayes NAS** (`tpe`): lightweight tree-structured Parzen estimator over discrete bake categoricals in the spirit of practical Bayesian optimization [26].

We also run **Ours (hybrid GA–PPO)** (`hybrid_lstm`): the multi-branch hybrid loop for 5,000 evaluations with LSTM/xLSTM in the block vocabulary and online reward-mode / fit–PPO hyperparameter co-tuning, as the proposed matched-budget arm. Results appear in Table 12 and Figures 6–7; healed-sample overlays in Figure 8.

Remark 2 (Trial complexity). Per outer iteration the dominant cost is $O(T_{\text{fit}} \cdot C_{\text{fwd}})$ for fit steps times one forward bake. Bake-width caps (tiny residual cells) keep C_{fwd} far below full-band Demucs/DiffWave forwards screened in Related Work.

3.9 Hyperparameters

Table 2 lists the CUDA search defaults used for the 5k-gate freeze reported in Results. Under the near-ceiling residual regime (Section 3.7), these outer-loop and fit hyperparameters are co-tuned with architecture; PBT and plateau adaptation continue to mutate several of them online. Canonical companion pseudocode lives in `docs/PSEUDOCODE.md` (mirrored Algorithms 1–9).

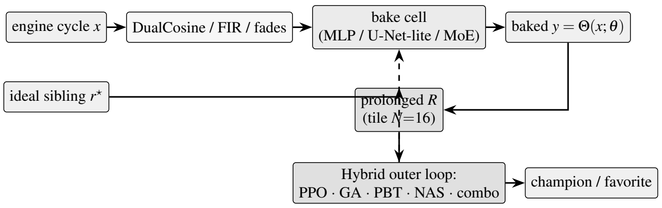

Figure 2: DenoiseOpt bake path and residual-scored hybrid meta-search. Classical seam ops and optional residual cells form a bake cell. Prolonged R ranks proposals. The outer loop rotates PPO/GA/PBT/NAS/combo branches. Dashed arrow: architecture proposals update the cell.

Table 2: Hybrid search hyperparameters (RTX 3090 DenoiseOpt CUDA search runner / `denoise_meta_evo.py` defaults). Co-tuned with reward shaping under near-ceiling R (Section 3.7).

Parameter	Value
Population size n	12
GA tournament k	3
GA crossover fraction p_c	0.5 (else clone elite parent)
GA mutation	≥ 1 mutate after inherit (+50% second mutate)
PPO clip ε	0.2 (default; searchable $\{0.1, 0.2, 0.25, 0.3\}$)
Adam learning rate (fit)	3×10^{-3} (default; PBT-searchable)
Adam learning rate (policy)	3×10^{-4}
Entropy coefficient (PPO)	0.02 (default; searchable $\approx$ 0.005–0.06)
PBT exploit / mutate	elite fraction 0.25; multi-step arch+hp mutate
MoE routing	`sequential` or `moe_parallel` when enabled
Plateau boredom threshold	1000 iters without champ improve
Cycle length L / tiles N / seam W	256 / 16 / 8
Fit steps / batch (default)	24 / 48
Algo tag	`PPO+GA+PBT+NAS+depth+MoE`

4 Experiments

Protocol freeze. Evaluation follows `paper/v7/EVAL_PROTOCOL.md` (v1, extended for peer-review strata). Primary metric: prolonged R. Secondary: SNR, SDR, wrap-jump (diagnostic), and required seam-local edge RMSE on tiled audio. Holdout seed `20260719`. Search seed `1902771841`. We report the 5k-gate freeze plus frozen holdout. We make no unfinished-budget claims. PESQ/STOI stay out of scope on non-speech cycles. LibriSpeech and MUSDB are not used. Venue positioning: DSP / synthesis artifact repair (DAFx / AES / arXiv cs.SD), not general speech-enhancement leaderboards.

4.1 Dataset

Creation. The paper-facing evaluation corpus is a frozen draw from the same synthetic generator used by the CUDA search runner (`make_batch` in `overnight_gpu_rl_arch.py`). Each cycle has length L=256. A two-harmonic sine is drawn with frequency $\sim \mathcal{U}(1,4)$ and phase $\sim \mathcal{U}(0,2\pi)$. An open-wrap seam cliff of amplitude $\pm\mathcal{U}(0.08, 0.43)$ is applied over `SEAM_W=8` samples at both ends. Seam-boosted Gaussian noise (scale 0.02 at the ends, 0.15× in the mid-cycle) is added to form the engine input. The ideal withholds the cliff and is used only as the scoring target $r^\star$ (never as a method output). No-bake leaves the cracked engine unchanged and scores that unrepaired x against the same $r^\star$. Prolonged residual R tiles each cycle N=16 times before the RMS ratio. The holdout seed is `20260719` (distinct from the hybrid search seed `1902771841`). The search campaign draws fresh i.i.d. batches from this generator and does not train on the frozen holdout tensors. There is no supervised clean/noisy training split of studio audio.

Multi-family diversity ($\geq$ 20 waveforms). Beyond the single sine+cliff holdout, Phase 3a scores methods on **20** generative draws: ten family variants (`sine_cliff`, `harmonic_fft`, `am_fm`, `nonlinear`, `combo`, `triple_mix`, `extreme_overlay`, `open_wrap_bias`, `soft_noise`, `wide_seam`) × two seeds (`20260719` and `20260720`), batch 64 each. These Python generative variants stress harmonic / AM–FM / nonlinear / overlay / open-wrap behavior of `make_batch` and remain the primary SOTA matrix. Separately, `export_sound_bench_tiles` exports **20** Rust `sound_bench` engine/ideal pairs (10 families × 2 seeds, L=256, byte-aligned with `generate_sound` / `generate_sound_ideal`) scored under the same residual geometry as a secondary transfer block (Table 13). Every method sees the same tensors per waveform. **Primary classical comparison** is absolute prolonged R against the non-AI board (no-bake, DualCosine, `seam_fir3`, poly, fades, VA residuals; Section 3.4). The search objective is maximize R toward the ideal sibling (best R=1). DualCosine is one classical comparator and a PPO centering constant; it is not the reward target.

Dataset metrics. A metrics draw of 4,096 cycles under the holdout seed yields mean ideal RMS 0.721 ± 0.016, mean engine residual RMS 0.021 ± 0.013, and mean absolute wrap jump $|x_0 - x_{L-1}|$ of 0.913 ± 0.634. No-bake (passthrough) on that draw scores mean $R \approx 0.971$. Method tables use a matched score batch of 64 cycles from the same seed (tensors under `brand/artifacts/canonical_eval_dataset/`), five consecutive seeds for mean±std on the classical table, and the 20-waveform matrix for SOTA secondary metrics. Figure 3 summarizes the 4,096-cycle metrics draw: ideal RMS clusters near 0.72, wrap jumps span nearly three orders of magnitude (median $\approx$ 0.83), and no-bake R is high on

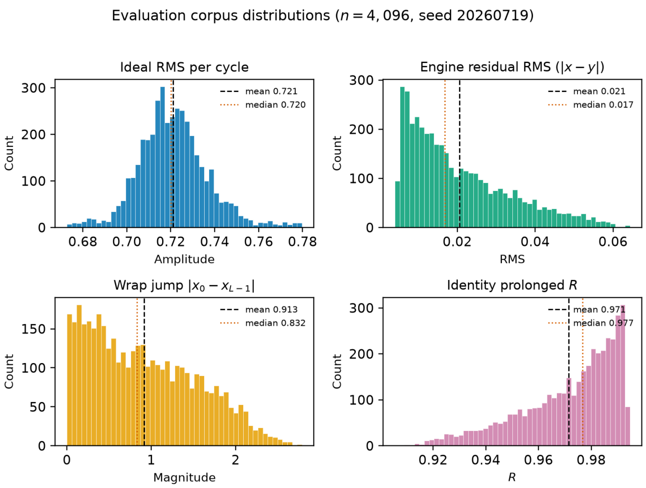

Figure 3: Evaluation corpus statistics on the frozen holdout seed (20260719, n=4,096 cycles). Top left: ideal RMS per cycle. Top right: engine residual RMS before any bake. Bottom left: wrap jump magnitude $|x_0 - x_{L-1}|$ (cliff hardness proxy). Bottom right: no-bake prolonged R (passthrough, $x \mapsto x$). Dashed and dotted vertical lines mark mean and median. Hard-cliff reporting (Section 5) uses the top 10% wrap-jump stratum where no-bake R drops.

easy tiles but falls below 0.91 on the hardest decile of wrap jumps. Figure 1 is the documented sine-wrap edge case: a high-$|wrap|$ holdout tile where cliff modulation is visible, scored with DualCosine and the top neural favorite under the same residual R (tile $R \approx 0.9917$, favorite batch mean $R \approx 0.991$).

Hybrid search campaign (5k clean gate). A CUDA search tagged `PPO+GA+PBT+NAS+depth+MoE` runs with a large declared iteration budget (order 10^5). At the reported freeze (run `gpu-rl-arch-20260719T083019Z`, $\approx$6,922 clean iters / $\approx$ 7.9 h on RTX 3090) the champion residual is $R \approx 0.99093$ against DualCosine baseline ≈ 0.81658 ($\Delta R \approx +0.174$). Branch-best residuals at the freeze: GA ≈ 0.99016, PBT ≈ 0.99012, PPO ≈ 0.98924, combo ≈ 0.98854, NAS ≈ 0.98677. Isolated short-budget re-runs (GA-only, PPO-only, GA+PPO, full, 150 iters, same search seed, plateau adapt off) are reported beside those freezes in Table 10. Figures in Section 5 regenerate from `history.jsonl` near this gate.

Inference, classical, and learned baselines. High-R fitted cells ($R \geq 0.98$) are timed on CUDA (batch 64, prolonged residual). Ten non-learnable bake denoisers share the frozen canonical holdout against the neural favorite (Table 4). Cycle-local BLIT/BLEP, PolyBLEP, and BLAMP seam repairs are scored on the same holdout / cliff / multi-family protocol (Table 9; `va_seam_blep.json`) and illustrated on the intro high-wrap tile (Figure 5). Fixed-arch MLP-on-R and CNN/U-Net-lite-on-R baselines (no outer NAS) are trained on $1-R$ with the same generator and scored in the SOTA matrix. Phase B adds a primary corrupt→corrupt Noise2Noise seam CNN (train seed `424242`, disjoint from holdout), a secondary sibling-supervised ceiling on the same architecture, and lightweight LSTM / depthwise 1D-CNN seq restorers (train seed `424243`). No search-champion weights initialize those baselines. Family-conditional hardness uses the separate Rust procedural 100,000-cycle bench reported in Results.

Cliff strata and wavetable-native realism. A 4,096-tile holdout draw (seed `20260719`) is stratified by engine wrap-jump percentiles (top 25% / top 10%; cutoffs frozen in `cliff_strata.json`). Hard-cliff is the operative regime for claims about DualCosine collapse vs favorite retention. Edge RMSE is the locked required seam-local secondary; click energy is optional. Wavetable-native transfer (Phase F1) scores true ReelSynth-exported factory periods (primary) and external AKWF CC0 WAVs (secondary) under the same close-seam→open-wrap protocol. Procedural stand-ins are tertiary smoke only. Speech/music LibriSpeech/MUSDB corpora remain off.

5 Results

Family-conditional hardness (completed bench). RQ: which procedural families dominate remaining residual mass? On the fitted DenoiseOpt 100,000-cycle bench, denoise score D is lowest for `nonlinear` ($D \approx 0.39$) and `combo` ($D \approx 0.40$), then `triple_mix` / `extreme_overlay` ($D \approx 0.51$–0.52), while `harmonic_fft` and `am_fm` sit near $D \approx 0.69$–0.70. Lit-combo champion family stress (prolonged residual) similarly ranks `open_wrap_bias` ($R \approx 0.83$) and `extreme_overlay` / `combo` ($R \approx 0.88$–0.89) below `harmonic_fft` ($R \approx 0.95$). Hard sounds recur.

Hybrid search (5k-gate freeze). RQ: what champion R does the hybrid search campaign reach at the clean 5k-iteration gate? At the reported 5k-gate search freeze ($\approx$ 6,922 clean iterations, $\approx$ 7.9 h on RTX 3090) the champion residual is $R \approx 0.99093$ with DualCosine ≈ 0.81658 on the CUDA search runner ($\Delta R \approx +0.174$). Monitoring plots below are regenerated from that freeze. Declared larger budgets continue. This paper reports the gate freeze only.

Inference time at matched score. RQ: among near-max-R fitted cells, which is fastest at matched score? Across diverse high-R fitted cells ($R \geq 0.98$), we timed CUDA forward passes (batch 64, prolonged residual). Within a tight band of the best residual ($R \geq R_{\max} - 5 \cdot 10^{-4}$), the **favorite** is the lowest-latency model: `champion_iter_000235` with $R \approx 0.9910$, $\approx$ 3.15 ms/batch, $\approx$ 37k parameters (versus a higher-capacity max-R cell at $R \approx 0.9914$ but $\approx$ 7.18 ms/batch / $\approx$ 104k params). Selection rule: maximize R, then minimize latency inside the band.

5.1 Top-5 architectures

RQ: which fitted bake graphs achieve near-max R at lowest latency on the inference bench? Table 3 ranks five distinct fitted bake graphs from the CUDA inference bench (`inference_bench.json`, batch 64), preferring residual

Table 3: Top-5 distinct fitted architectures on the CUDA inference bench (batch 64, prolonged R, seed geometry matching the search campaign). Favorite is latency-best inside the near-max-R band ($R \geq R_{\max} - 5 \cdot 10^{-4}$).

Tag	R_{saved}	R_{live}	ms/batch	params
`iter_000157`	0.9914	0.9913	7.18	104 484
`iter_000205`	0.9911	0.9907	7.29	70 562
`iter_000235*`	0.9910	0.9915	3.15	37 489
`iter_000397`	0.9910	0.9907	6.80	97 262
`iter_000229`	0.9909	0.9902	6.67	100 500

*Favorite (fastest in $R \geq R_{\max} - 5 \cdot 10^{-4}$ band).

Table 4: Method scores on the canonical sine+cliff holdout (batch 64, seed `20260719`, L=256, N=16). Mean±std over five consecutive seeds. Every method is scored vs the same ideal sibling $r^\star$. No-bake = unrepaired engine (not $r^\star$). Rank by absolute R against the classical board; DualCosine is one classical row, not the sole comparator. Primary metric is prolonged R (1 = best).

Method	R	R (5-seed)	ms/batch
neural favorite	0.9911	0.9912 ± 0.0002	2.56
no-bake (passthrough)	0.9718	0.9726 ± 0.0021	0.31
`seam_fir3`	0.9610	0.9616 ± 0.0019	0.67
classic quadratic	0.8449	0.8453 ± 0.0117	1.59
DualCosine	0.8249	0.8258 ± 0.0137	1.35
cosine fade	0.8249	0.8258 ± 0.0137	1.59
crossfade	0.8116	0.8130 ± 0.0145	1.50
hann blend	0.8048	0.8060 ± 0.0156	0.54
soft wide fade	0.7432	0.7378 ± 0.0171	2.64
detrend	0.4079	0.4091 ± 0.0386	0.36
ensemble detrend+DC+FIR	0.4024	0.4027 ± 0.0387	1.56

R first and latency on ties, and skipping near-duplicate copies of the same topology signature.

Classical vs favorite (canonical holdout). Table 4 scores the classical (non-AI) bake set—including no-bake, DualCosine, and `seam_fir3`—plus the neural favorite on the frozen corpus of Section 4.1 (CUDA, batch 64, seed `20260719`). Every row uses the same ideal sibling $r^\star$; higher R means closer to that sibling. DualCosine reaches $R \approx 0.825$ ($\approx$ 1.35 ms/batch), near the search-monitor DualCosine ≈ 0.817 on live draws from the same generator. The best *active* classical repair is seam-local 3-tap FIR (`seam_fir3`) at $R \approx 0.961$ / ≈ 0.67 ms/batch (0.962 ± 0.002 over five seeds). No-bake (passthrough; unrepaired engine vs $r^\star$) scores $R \approx 0.972$ and is a do-nothing reference, not the ideal. The neural favorite reaches $R \approx 0.991$ (0.991 ± 0.0002 over five seeds) at ≈ 2.56 ms/batch ($\approx$ 37k params): $\Delta R \approx +0.019$ vs no-bake, $\Delta R \approx +0.030$ vs `seam_fir3`, and $\Delta R \approx +0.166$ vs DualCosine.

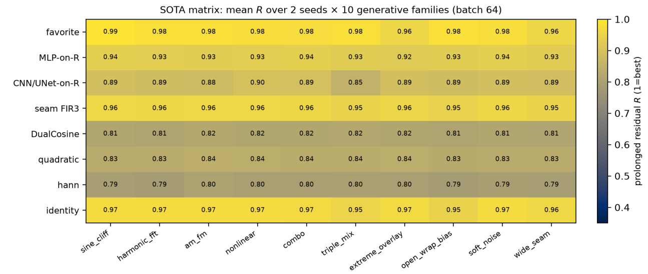

Figure 4: Colorblind-safe (`cividis`) heatmap of mean prolonged R for selected methods × generative families (2 seeds each). Higher is better.

5.2 SOTA matrix (multi-family + secondary metrics)

RQ: does the meta favorite beat the classical non-AI board (no-bake, DualCosine, FIR, poly, fades) and fixed MLP/CNN baselines across twenty generative families? Table 5 reports mean±std over the 20-waveform multi-family set (Section 4.1), with canonical-holdout SNR/SDR/wrap-jump for the same method set from `sota_matrix.json`. PESQ/STOI/MUSHRA are omitted for non-speech cycles (Limitations). Primary reading is absolute R rank vs classical rows (no-bake 0.965, poly 0.966, `seam_fir3` 0.955, DualCosine 0.814). Fixed MLP-on-R and CNN/U-Net-lite-on-R (no outer NAS) sit between DualCosine and the meta favorite. Paired Wilcoxon signed-rank (normal approx.) of favorite vs DualCosine on the 20 waveforms: W=0, $p \approx 8.9 \times 10^{-5}$, mean ΔR=+0.162 (bootstrap 95% CI $[0.155, 0.170]$); vs no-bake the multifamily mean gap is smaller ($\approx +0.012$) but still favors the searched bake on this generator.

5.3 Hard-cliff strata

RQ: does high no-bake R hide DualCosine failure on large wrap jumps? Table 6 stratifies a 4,096-tile holdout draw (seed `20260719`) by engine wrap-jump percentiles (p75$\approx$ 1.387, p90$\approx$ 1.832; frozen in `cliff_strata.json`). No-bake R barely moves (all 0.9715 $\to$ top-10% 0.9667) because residual mass remains small relative to ideal RMS under (1). DualCosine collapses on hard cliffs (0.819 $\to$ 0.657), while the meta favorite holds ($R \approx 0.9915$). Favorite wrap-jump on the top-10% subset stays ≈ 1.83 (vs no-bake 2.09). The claim is tiled residual repair, not endpoint-zeroing. **Edge RMSE** is the locked required seam-local secondary (EVAL_PROTOCOL): on the top-10% subset it drops from no-bake 0.095 to favorite 0.021 (DualCosine 0.990). Click energy remains an optional diagnostic.

5.4 N2N and seq baselines

RQ: how does DenoiseOpt compare to true Noise2Noise-style and lightweight seq restorers on the same generator? Primary corrupt$\to$corrupt N2N (20k Adam steps, 53k params, train seed `424242`) reaches canonical $R \approx 0.9917$, matching the meta favorite within ≈ 0.001. Secondary sibling-supervised N2N reaches $R \approx 0.9933$. A bidirectional LSTM (76k params, 15k steps, train seed `424243`) is the supervised ceiling at

Table 5: SOTA comparison on 20 generative waveforms (10 families × 2 seeds, batch 64). R/SNR/SDR/wrap-jump are multi-family mean±std. Rank by absolute R vs the classical non-AI board (no-bake, poly, FIR, DualCosine, ...). Column ΔR* is vs DualCosine only (one classical gap for reporting; canonical freeze seed `20260719`); ms* likewise. The search objective is maximize absolute R toward the ideal sibling, not DualCosine matching. Modern baselines (N2N/seq/poly) use disjoint train seeds (`424242`/`424243`); poly is classical (no train). Primary metric R (1 = best; closer to ideal sibling). No PESQ/STOI/MUSHRA.

Method	R (20-wave)	SNR (dB)	SDR (dB)	wrap-jump	ms*	params	ΔR*
neural favorite	0.977 ± 0.010	35.0 ± 3.2	35.1 ± 3.2	0.90 ± 0.14	2.44	37 489	+0.166
N2N corrupt→corrupt	0.970 ± 0.021	33.9 ± 4.1	34.0 ± 4.1	0.90 ± 0.15	–	53 506	+0.167
seq CNN1D	0.969 ± 0.016	32.2 ± 3.5	32.4 ± 3.5	0.92 ± 0.15	–	3 242	+0.162
N2N sibling-sup.	0.969 ± 0.022	33.7 ± 4.3	33.8 ± 4.3	0.91 ± 0.15	–	53 506	+0.168
poly seam (d=3)	0.966 ± 0.008	31.2 ± 2.1	31.3 ± 2.1	0.91 ± 0.15	1.47	0	+0.148
no-bake	0.965 ± 0.008	30.9 ± 1.9	31.0 ± 1.9	0.91 ± 0.15	0.00	0	+0.147
seq LSTM	0.956 ± 0.037	32.2 ± 5.8	32.3 ± 5.8	0.86 ± 0.15	–	75 746	+0.170
`seam_fir3`	0.955 ± 0.005	28.1 ± 1.2	28.1 ± 1.2	0.46 ± 0.08	0.59	0	+0.136
MLP-on-R	0.932 ± 0.006	24.6 ± 0.9	24.6 ± 0.9	0.64 ± 0.07	2.76	8 604	+0.113
CNN/U-Net-on-R	0.886 ± 0.012	19.5 ± 0.9	19.5 ± 0.9	0.48 ± 0.06	5.01	11 752	+0.069
classic quadratic	0.835 ± 0.012	17.9 ± 1.1	17.7 ± 1.2	0.32 ± 0.05	1.78	0	+0.020
DualCosine	0.814 ± 0.013	16.9 ± 1.2	16.8 ± 1.2	0.32 ± 0.05	1.48	0	0
cosine fade	0.814 ± 0.013	16.9 ± 1.2	16.8 ± 1.2	0.32 ± 0.05	2.64	0	0
crossfade	0.802 ± 0.013	16.7 ± 1.2	16.4 ± 1.2	0.91 ± 0.15	1.70	0	−0.013
hann blend	0.793 ± 0.015	16.1 ± 1.2	15.8 ± 1.2	0.32 ± 0.05	0.57	0	−0.020
soft wide fade	0.729 ± 0.016	14.0 ± 1.1	13.6 ± 1.1	0.63 ± 0.08	3.69	0	−0.082
detrend	0.375 ± 0.040	5.5 ± 1.0	5.4 ± 1.0	0.00 ± 0.00	0.12	0	−0.417
ensemble DC+FIR	0.368 ± 0.040	5.1 ± 1.0	5.0 ± 1.0	0.12 ± 0.02	1.99	0	−0.422

Table 6: Cliff strata on 4,096 holdout tiles (seed `20260719`). Strata by engine wrap-jump. Primary R (1 = best) and mean wrap-jump after bake. N2N primary = corrupt→corrupt; N2N secondary = sibling-supervised ceiling; seq = engine→ideal.

Method	all R	all jump	top25 R	top25 jump	top10 R	top10 jump
seq LSTM (ceiling)	0.9953	0.865	0.9955	1.637	0.9955	1.810
N2N sibling-sup.	0.9936	0.879	0.9936	1.668	0.9934	1.842
N2N corrupt→corrupt	0.9922	0.870	0.9924	1.647	0.9922	1.821
neural favorite	0.9911	0.869	0.9915	1.645	0.9915	1.825
seq CNN1D	0.9863	0.881	0.9869	1.684	0.9867	1.871
no-bake	0.9715	0.913	0.9713	1.797	0.9667	2.094
`seam_fir3`	0.9611	0.456	0.9461	0.893	0.9429	1.034
DualCosine	0.8193	0.335	0.6864	0.290	0.6574	0.335

n: all 4096, top25 1024, top10 410.

$R \approx 0.9952$. A tiny depthwise 1D CNN (3.2k params) reaches $R \approx 0.9864$. Holdout tiles never enter training. Classical DualCosine / `seam_fir3` remain in Table 6.

5.5 Wavetable-native transfer

RQ: does the wrap protocol transfer beyond synthetic sine cliffs onto *true* exports and third-party OA cycles? Table 8 scores ReelSynth-exported factory periods (primary realism, n=25; byte-true factory frames resampled to L=256) and Adventure Kid Waveforms (AKWF) CC0 single-cycle WAVs (secondary, n=24) after close-seam idealization and the same open-wrap cliff. Procedural stand-ins are demoted to tertiary smoke and no longer carry the primary realism claim. On exported factory periods the favorite ($R \approx 0.911$) trails no-bake / `seam_fir3` / poly (domain shift) while still beating DualCosine (0.883). On AKWF OA cycles the favorite ($R \approx 0.970$) matches no-bake and beats DualCosine (0.936). LibriSpeech/MUSDB are not used.

5.6 Polynomial baseline and jump-aware control

RQ: does a cheap classical poly fitter close the S7 residual, and does endpoint pinning show the wrap-jump vs R tradeoff? A seam-local degree-3 polynomial fitter (no learning) reaches canonical $R \approx 0.972$ ($\approx$ no-bake) with wrap-jump essentially unchanged (`poly_baseline.json`). SSM seq models are deferred; the LSTM remains the supervised seq ceiling. An endpoint-pin control (`jump_control.json`) drives wrap-jump to 0 on all / hard subsets while dropping R (all 0.906; hard 0.822). Methods that zero wrap-jump are not the same

Table 7: Required seam-local secondary on top-10% wrap-jump tiles (n=410): edge RMSE (lower better). Frozen in `cliff_strata.json`.

Method	top10 edge RMSE
seq LSTM	0.011
N2N sibling-sup.	0.018
neural favorite	0.021
N2N corrupt→corrupt	0.021
seq CNN1D	0.035
no-bake	0.095
`seam_fir3`	0.164
DualCosine	0.990

Table 8: Wavetable-native wrap protocol (Phase F1). Close-seam ideal → open-wrap cliff (seed `20260719`). Mean prolonged R. Primary = ReelSynth export; secondary = AKWF CC0 files.

Method	Export (n=25)	AKWF OA (n=24)
endpoint-pin (mean)	0.939	0.978
`seam_fir3`	0.971	0.975
poly seam (d=3)	0.967	0.970
no-bake	0.967	0.969
neural favorite	0.911	0.970
DualCosine	0.883	0.936
N2N corrupt→corrupt	0.869	0.974
seq CNN1D	0.839	0.962
seq LSTM	0.769	0.948

objective as prolonged R; we do not claim wrap-jump SOTA for the favorite.

5.7 Classical VA seam repairs (BLIT/BLEP, PolyBLEP, BLAMP)

RQ: do the cited VA antialias tools [1–3] beat DualCosine on prolonged R when applied as cycle-local seam residuals?

What each operator changes. Figure 5 shows the family on one high-wrap sine+cliff tile (same seed and tile as Figure 1). **BLIT/BLEP** (Stilson/Smith lineage) treats the wrap as a value step. It subtracts a band-limited step residual (here a raised-cosine BLEP-family lobe scaled by the measured wrap jump) so the tiled click spectrum is softened near the seam. **PolyBLEP** (Nam et al.) is a cheap polynomial surrogate of that step residual with fractional-delay support near phase wrap. It edits only a few samples on each side of the discontinuity. **BLAMP** (Esqueda et al.) corrects a slope or corner jump (discontinuity in the first derivative), which matters for triangles, rectification, and hard clipping. On a pure value cliff the slope jump is small, so BLAMP is nearly a no-bake passthrough. **DualCosine** (our classical bake baseline) fades both ends of the stored period with raised-cosine windows. It does not insert a BLEP-style residual. It trades wrap energy for a wide end-fade distortion that prolonged R penalizes on hard cliffs.

Scores. Table 9 reports batch means from `va_seam_blep.json` (seed `20260719`, batch 64). Implementation: `reelsynth/scripts/baselines/va_seam_blep.py` (PolyBLEP matches `osc::va::poly_blep`; BLIT/BLEP uses a raised-cosine step residual; BLAMP applies a poly-BLAMP lobe to the wrap slope jump). PolyBLEP reaches mean $R \approx 0.929$ ($\Delta R \approx +0.104$ vs DualCosine) but trails no-bake / `seam_fir3` / the meta favorite. BLIT/BLEP nearly zeros wrap-jump (0.004) while dropping mean R to 0.868 and collapsing on hard cliffs ($R \approx 0.712$). BLAMP is near-no-bake on this value-cliff geometry (mean $R \approx 0.972$). On the annotated severe tile in Figure 5, tile R is lower still (PolyBLEP ≈ 0.869, BLIT/BLEP ≈ 0.703, DualCosine ≈ 0.603, BLAMP $\approx$ engine), which matches the hard-cliff strata. Honest reading: these classical tools are now scored, not only cited. They beat DualCosine on mean R in places. None matches the favorite on prolonged R.

5.8 Transfer conditions

RQ: when does hybrid search win vs the classical non-AI board (no-bake / FIR / DualCosine) and supervised nets? Per-cycle ΔR on export, AKWF, Rust `sound_bench`, and sine+cliff holdout (`transfer_failures.json`): favorite win-rate vs DualCosine is high on sine cliffs (1.00) and AKWF (0.88), and moderate on export (0.64) / Rust (0.75); win-rate vs no-bake is high only on sine cliffs (0.92) and collapses on export (0.24) / Rust (0.20). Search value is therefore: hard cliffs, OA tracks where DualCosine collapses, and compact bake graphs without engine→ideal labels at search time. No-bake/FIR win on mild seams and OOD factory geometry (honest domain shift, not buried).

5.9 Ablations and compute

RQ: how do search branch-best freezes compare to short isolated GA/PPO budgets, and what compute was used? Table 10 reports two views. Search *branch-best freezes* at the 5k gate come from one hybrid history. Isolated short-budget re-runs (GA-only, PPO-only, GA+PPO, full) use 150 iters on the same search seed with plateau adapt off. Short-budget champions sit below the multi-hour freeze by construction. They are labeled as such. Table 11 summarizes the search-campaign compute budget. Peak VRAM is from an `nvidia-smi` poll during a short full-branch search replay (30 iters) plus a champion forward+FitCell replay. The campaign `history.jsonl` did not log memory.

5.10 Meta-learning approach comparison (5k matched budget)

RQ: under a matched 5,000-evaluation budget with LSTM and xLSTM in the searchable bake vocabulary, how do Random NAS, Cont. CMA-ES, Arch REINFORCE, Aging evolution, and TPE Bayes NAS compare to **Ours (hybrid GA–PPO)**? Table 12 reports champion prolonged R, ΔR vs DualCosine, wall hours, and whether the champion graph used an LSTM or xLSTM block. At the completed gate, **Ours** leads on ab-

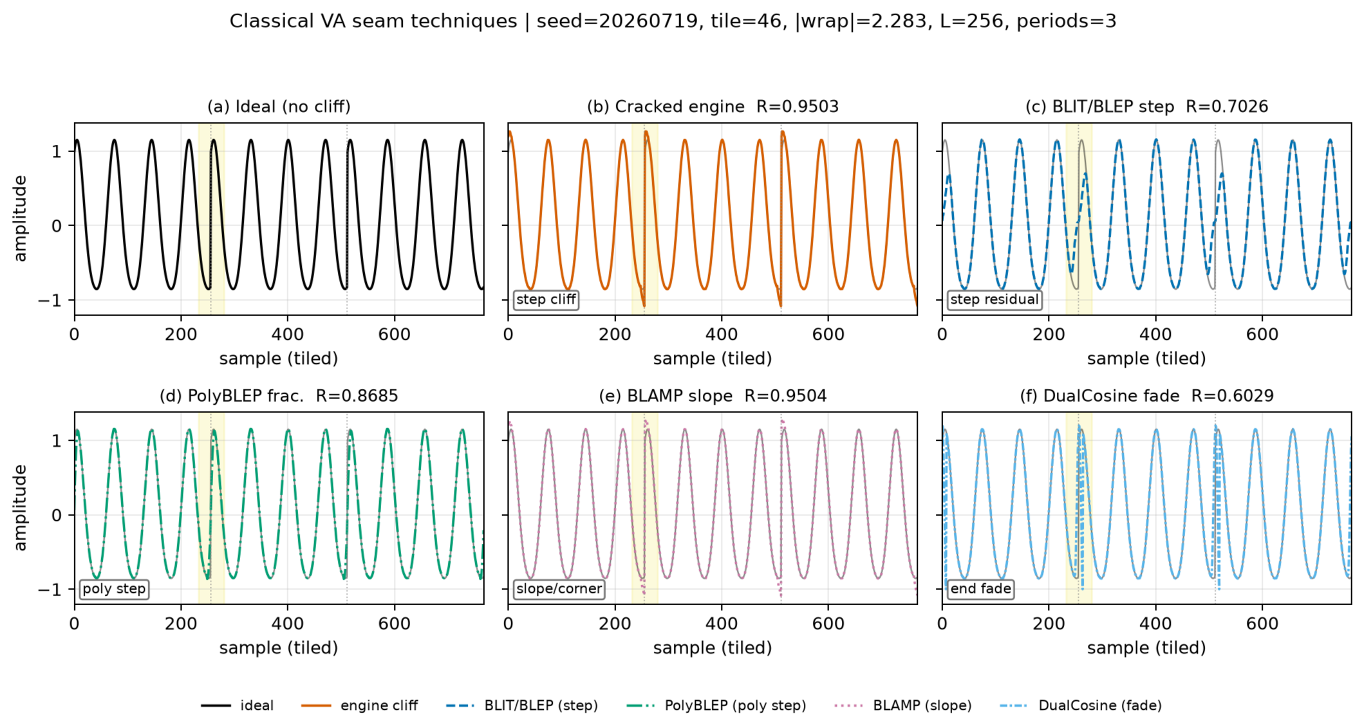

Figure 5: Classical VA seam techniques on one cracked sine+cliff tile (seed `20260719`, tile 46, $|wrap|$=2.283, L=256, three tiled periods). Yellow band marks the first seam neighborhood. Panel tile R (1 = best): (a) ideal (unscored sibling). (b) cracked engine R=0.9503. (c) BLIT/BLEP step residual R=0.7026. (d) PolyBLEP fractional-delay residual R=0.8685. (e) BLAMP slope/corner residual R=0.9504 (near no-bake on value cliffs). (f) DualCosine end fade R=0.6029. Colorblind-safe hues with distinct linestyles for B&W print. Batch means appear in Table 9. Accessibility: six-panel comparison of ideal, cracked, BLIT/BLEP, PolyBLEP, BLAMP, and DualCosine.

Table 9: Classical VA seam repairs on the residual protocol (seed `20260719`). Canonical holdout: R, SNR/SDR, wrap-jump, edge RMSE. Cliff top-10%: R (n=410). Multi: mean R over 20 generative waveforms.

Method	R	SNR	SDR	jump	edge	top10 R	multi R
BLAMP	0.9718	32.6	32.7	0.873	0.080	0.9667	0.965
no-bake	0.9718	32.6	32.7	0.873	0.080	0.9667	0.965
`seam_fir3`	0.9610	29.6	29.6	0.438	0.112	0.9429	0.955
PolyBLEP	0.9288	25.9	25.8	0.102	0.205	0.8641	0.924
BLIT/BLEP	0.8680	20.4	20.3	0.004	0.365	0.7116	0.860
DualCosine	0.8249	18.0	17.8	0.292	0.506	0.6574	0.814

Latency (canonical, CUDA): PolyBLEP ≈ 1.43 ms/batch; DualCosine ≈ 1.06; BLIT/BLEP ≈ 123; BLAMP ≈ 54. Frozen in `va_seam_blep.json`.

solute champion R ($R \approx 0.99146$, $\Delta R \approx +0.17488$ vs DualCosine ≈ 0.81658; ≈ 8.46 h), with Aging evolution second ($R \approx 0.99079$). Neither Ours nor Aging evolution selected an LSTM block in the champion graph; Arch REINFORCE / Aging evolution champions used xLSTM, and TPE used LSTM (vocab presence, not a full-stack claim). Figure 6 is the at-a-glance bar comparison; Figure 7 shows champion-R learning curves (x-axis capped at 5,000). Figure 8 shows a holdout wrap-seam heal (seed `20260719`) for the refit Ours champion against no-bake, DualCosine, and ideal sibling $r^{\star}$. Source: `meta_approach_compare.json` (search seed `1902771841`).

5.11 Rust `sound_bench` transfer (20 tiles)

RQ: does the Python-trained favorite transfer to Rust `sound_bench` tiles against the classical non-AI board? Table 13 scores no-bake, `seam_fir3`, DualCosine, other classical rows, and the Python-trained neural favorite on 20 Rust-exported tiles (10 families × 2 seeds). This closes the prior gap where multi-family tables used only Python generative stand-ins. Honest transfer note: the favorite was searched on Python `make_batch` sine+cliff geometry. On Rust families it still beats DualCosine (mean ΔR+0.063, Wilcoxon p≈0.009) but trails no-bake and `seam_fir3` on absolute

Table 10: Ablations. Left block: search branch-best freeze (run `gpu-rl-arch-20260719T083019Z`, final iter ≈ 6,922). Right block: isolated short-budget re-runs (150 iters, seed `1902771841`, plateau adapt off). R is prolonged residual on the runner geometry.

Config	Search freeze R	Isolated 150-it R
Full hybrid	0.99093	0.98113
GA-only	0.99016	0.98062
PBT (branch best / n/a isolated)	0.99012	—
PPO-only	0.98924	0.97932
GA+PPO	—	0.98007
combo (branch best)	0.98854	—
NAS (branch best)	0.98677	—
DualCosine (runner)	0.81658	0.81658

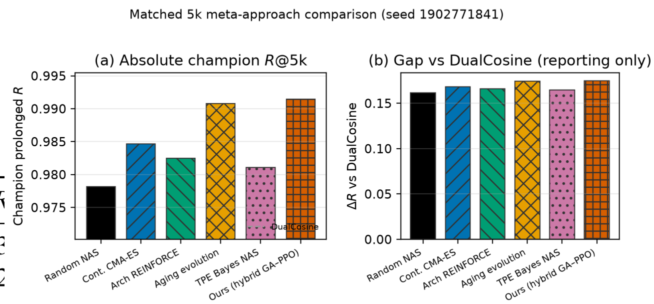

Figure 6: Matched 5k-budget meta-approach comparison (search seed `1902771841`): (a) champion prolonged R; (b) ΔR vs DualCosine (reporting gap only—objective is maximize absolute R toward the ideal sibling). Manuscript names; Okabe–Ito colors with distinct hatches for grayscale print.

Table 11: Compute budget for the reported 5k-gate search freeze (RTX 3090, PyTorch 2.6+cu124, search seed `1902771841`). Arch evaluations ≈ clean iterations.

Item	Value
GPU	NVIDIA GeForce RTX 3090
Elapsed wall time	≈ 7.92 h
Clean iterations / arch evals	≈ 6,922
Population size	12
Final champion R	0.99093
DualCosine baseline (runner)	0.81658
Peak VRAM (nvidia-smi replay)	≈ 3.29 GiB

Peak memory: max `nvidia-smi memory.used` during a 30-iter full-branch search probe (≈ 3.29 GiB) and champion forward+FitCell replay (≈ 3.17 GiB). The multi-hour search process itself left no memory log. Idle GPU occupancy was already ≈ 3.0 GiB from other sessions.

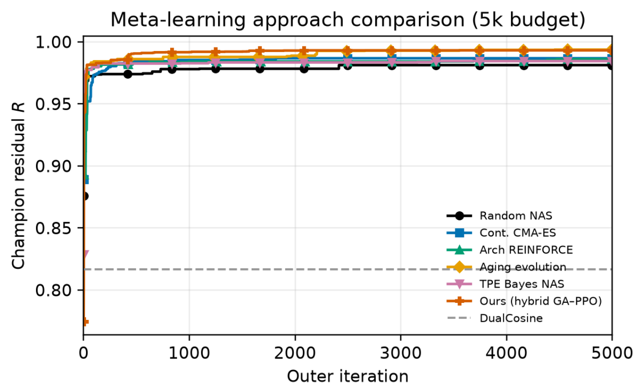

Figure 7: Matched 5k-budget meta-learning comparison: champion residual R versus outer iteration (x-axis capped at 5,000). DualCosine dashed as one classical comparator only—objective is maximize absolute R toward the ideal sibling; rank vs the classical board in Table 4. Accessibility: Okabe–Ito colors with distinct markers for grayscale print.

Table 12: Matched 5k-budget outer-loop comparison (search seed `1902771841`; LSTM+xLSTM in bake vocab; reward-mode + HP co-tune in Ours). Champion R vs ideal sibling; ΔR vs DualCosine is one classical gap only. Wall-h: per-method CUDA wall time. LSTM?/xLSTM?: whether the champion graph included that block.

Method	Champ R@5k	ΔR vs DualCosine	Wall-h	LSTM?	xLSTM?
Random NAS	0.97816	+0.16157	5.72	no	no
Cont. CMA-ES	0.98468	+0.16810	2.28	no	no
Arch REINFORCE	0.98247	+0.16588	25.93	no	yes
Aging evolution	0.99079	+0.17421	15.02	no	yes
TPE Bayes NAS	0.98106	+0.16448	11.18	yes	no
Ours (hybrid GA–PPO)	0.99146	+0.17488	8.46	no	no

Table 13: Secondary transfer on 20 Rust `sound_bench` tiles (byte-aligned export). Mean±std prolonged R vs the same ideal sibling. Rank by absolute R against classical non-AI rows; ΔR column is vs DualCosine only. Favorite beats DualCosine (mean ΔR+0.063, p≈0.009) but trails no-bake / FIR.

Method	R (20 Rust tiles)	ΔR vs DualCosine
no-bake	0.928 ± 0.084	+0.166
`seam_fir3`	0.896 ± 0.077	+0.134
neural favorite	0.825 ± 0.121	+0.063
classic quadratic	0.788 ± 0.110	+0.026
DualCosine / cosine fade	0.762 ± 0.118	0

R. The primary SOTA claim remains the Python multi-family matrix.

6 Discussion

Where residual mass remains. Across the 100,000-cycle procedural bench and lit-combo family stress tests, wrap error is not uniform. The same generator families repeatedly dominate the residual budget: `nonlinear`, `combo`, `triple_mix`, and `extreme_overlay` show the weakest denoise / residual scores, while `harmonic_fft` and `am_fm` are comparatively easy. `open_wrap_bias` can look strong on wrap-energy proxies yet still lag on prolonged residual.

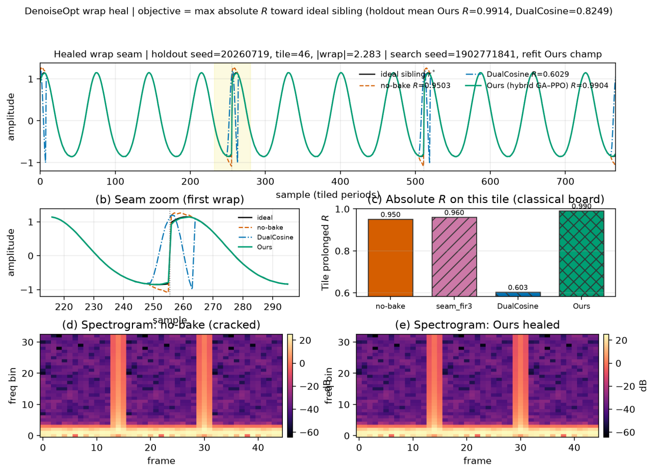

Figure 8: Healed wrap-seam example for the matched-suite **Ours (hybrid GA–PPO)** champion (arch+HP from `hybrid_lstm` summary; FitCell refit with search seed `1902771841`). Holdout tile (seed `20260719`): prolonged waveforms vs ideal sibling, seam zoom, per-tile absolute R vs no-bake / FIR / DualCosine, and spectrograms before/after heal. Cell weights are refit for visualization (suite checkpoints store arch+HP, not fitted `state_dict`); holdout mean R accompanies the tile view. DualCosine ΔR is reporting only.

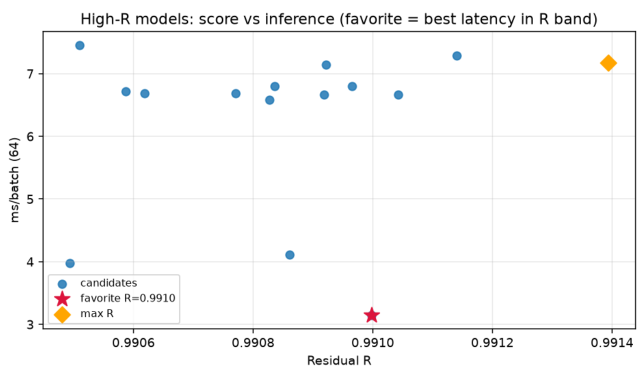

Figure 9: High-R fitted cells on CUDA (batch 64): prolonged residual R versus forward latency (ms/batch). Star marks the favorite (fastest inside the near-max-R band).

Cliff reduction and ideal-matched tiling are related objectives with different rankings.

No-bake R and hard-cliff honesty. Canonical no-bake $R \approx 0.97$ looks "easy" only if one treats R as a perceptual wrap score or confuses no-bake with the ideal sibling. No-bake is the unrepaired engine scored against $r^\star$; the ideal withholds the cliff and is never a method output. Under (1), small residual RMS relative to ideal RMS yields high R even when $|x_0 - x_{L-1}|$ is large. That same near-ceiling regime is why tiny ΔR still matters operationally: climbing $0.97 \to 0.991$ is most of the remaining residual budget, yet raw score differences are small enough that reward shaping and outer-loop hyperparameters must be tuned with the architecture (Section 3.7). Hard-

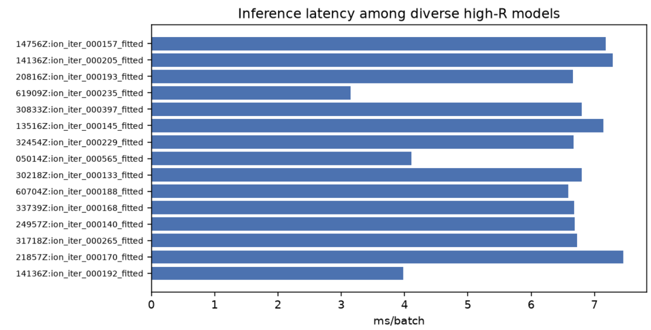

Figure 10: Inference latency (ms/batch, CUDA, batch 64) for diverse high-R fitted models from the inference bench.

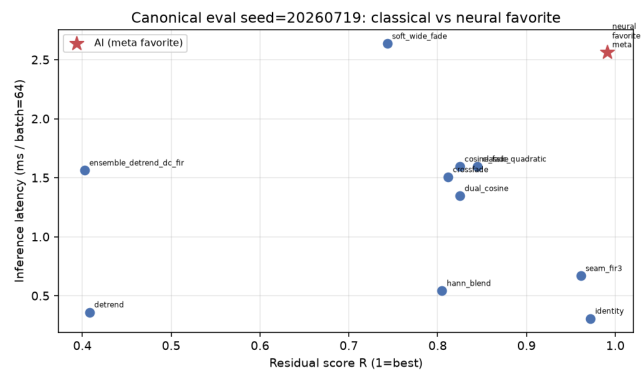

Figure 11: Canonical holdout (seed `20260719`, batch 64): classical bake methods and neural favorite in the R vs CUDA ms/batch plane. Star: favorite.

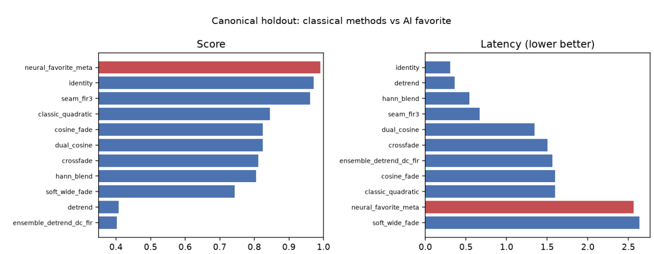

Figure 12: Canonical holdout ranking on prolonged residual R (left) and CUDA latency (right). DualCosine $R \approx 0.825$. Favorite highlighted.

cliff strata make the operative regime explicit: DualCosine drops to $R \approx 0.657$ on the top-10% wrap-jump tiles while the favorite stays near 0.9915. Favorite wrap-jump barely moves toward zero on that subset. We claim tiled residual repair toward $r^\star$, not endpoint closure.

N2N stress test and transfer conditions. On the sine+cliff holdout, corrupt→corrupt N2N matches the meta favorite within $\approx 10^{-3}$. Sibling-supervised N2N and the LSTM ceiling sit slightly above. That is expected for oracle-label training on the same generator and does not invalidate the hybrid search story: DenoiseOpt discovers compact bake graphs

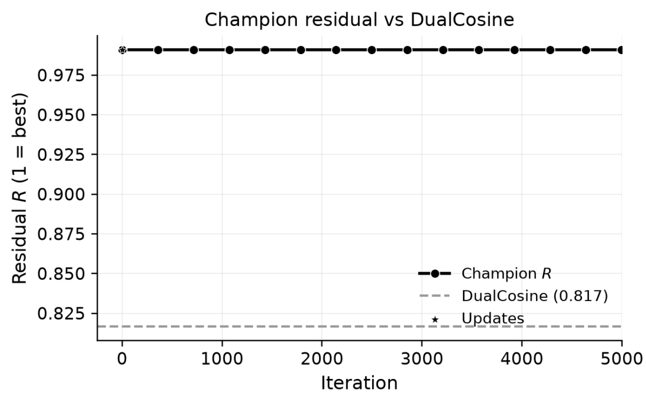

Figure 13: 5k-gate search freeze: champion residual R versus clean search iteration (x-axis capped at 5,000), with DualCosine baseline on the runner geometry (RTX 3090). Accessibility: black line with circle markers; DualCosine as a dashed gray baseline.

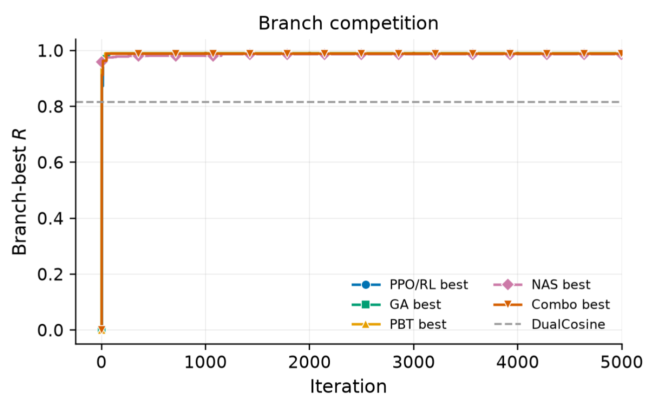

Figure 14: 5k-gate search freeze: per-branch best residual trajectories (PPO/RL, GA, PBT, NAS, combo) versus iteration (x-axis capped at 5,000). Accessibility: colorblind-safe colors plus distinct markers (circle, square, triangle, diamond, inverted triangle) for grayscale print.

without engine→ideal supervision at search time, and the hard-cliff DualCosine collapse contrast remains. Win conditions (`transfer_failures.json`): favorite beats DualCosine on sine cliffs and most AKWF / Rust tiles; no-bake/FIR lead on exported factory geometry (domain shift, not buried). On true ReelSynth exports the searched favorite trails no-bake and FIR; we report that gap rather than claim universal factory superiority.

Wrap-jump vs edge RMSE. Primary claim remains prolonged R. Wrap-jump is diagnostic only. Edge RMSE is the locked seam-local secondary: favorite reduces it on hard cliffs even when wrap-jump stays flat. Endpoint-pin zeros wrap-jump while hurting R, confirming the objectives differ.

Implication for this paper. Mean R (and mean DualCosine gaps) therefore compress a structured difficulty map. Reporting only the search-campaign mean on synthetic sine cliffs understates how much of the remaining $\sim$1% sits in a few hard families, and how DualCosine fails when wrap-jump is large. Hard-cliff strata and wavetable-native transfer (true exports + AKWF) are first-class empirical claims of the present work. The contribution is aimed at DSP / synthesis artifact repair (DAFx / AES / arXiv cs.SD), not general speech-enhancement leaderboards.

Hybrid search vs. plateau. Search traces at the 5k gate show early high champions then long plateaus near $R \approx 0.99$. That plateau is expected under near-ceiling R: further gains are small in absolute units and sensitive to reward shaping, entropy/clip, mutation rates, and plateau-triggered branch reweighting. Plateau adaptation (deeper cells, denser MoE graphs, higher GA/PBT weight) is the operational response inside one campaign. It does not by itself rebalance family-conditional risk. A natural next paper is family-conditional and cliff-conditional meta-learning: train or select denoisers per family, or condition the outer loop on measured wrap statistics, and publish per-family residual curves beside the scalar champion—together with ablations of reward shaping under the same compressed-R regime.

Hybrid branches. At the reported 5k-gate search freeze, GA and PBT lead branch-best residuals, with PPO close behind and NAS trailing on this seed (Table 10). These per-branch readings are snapshots from a single hybrid run, not causally isolated single-branch experiments. Isolated short-budget re-runs (150 iterations, same search seed) confirm relative ordering under a fixed compute cap (Table 10). The matched 5k outer-loop comparison (Table 12, Figures 6–7) places Random NAS, Cont. CMA-ES, Arch REINFORCE, Aging evolution, and TPE Bayes NAS beside **Ours (hybrid GA–PPO)** under a shared seed and bake vocabulary that includes searchable LSTM/xLSTM blocks. Ours wins the absolute-R gate ($R \approx 0.99146$) with Aging evolution close behind ($R \approx 0.99079$); Cont. CMA-ES is the wall-time leader (≈ 2.3 h) at lower champion R. Figure 8 shows the wrap-seam heal qualitatively on holdout geometry. The multi-family SOTA matrix (Table 5) shows the meta favorite ahead of DualCosine on R, SNR, and SDR across twenty Python generative draws, and ahead of no-bake / FIR on absolute R on that same generator. Fixed MLP/CNN-on-R baselines beat DualCosine and trail the searched favorite. On Rust `sound_bench` tiles (Table 13) the same favorite beats DualCosine (mean $\Delta R = +0.063$) and trails no-bake / FIR. That transfer gap matches the narrow Python training geometry.

7 Tooling and AI use

This section discloses research and writing tools, including AI systems, with scope and intent.

Literature discovery. Open literature was gathered with the Klaut Research Gateway (local OpenAlex / Crossref / arXiv providers) under queries spanning virtual-analog anti-aliasing, unsupervised restoration, HPO/PBT, RL–GA hybrids for NAS, and audio denoise architectures.

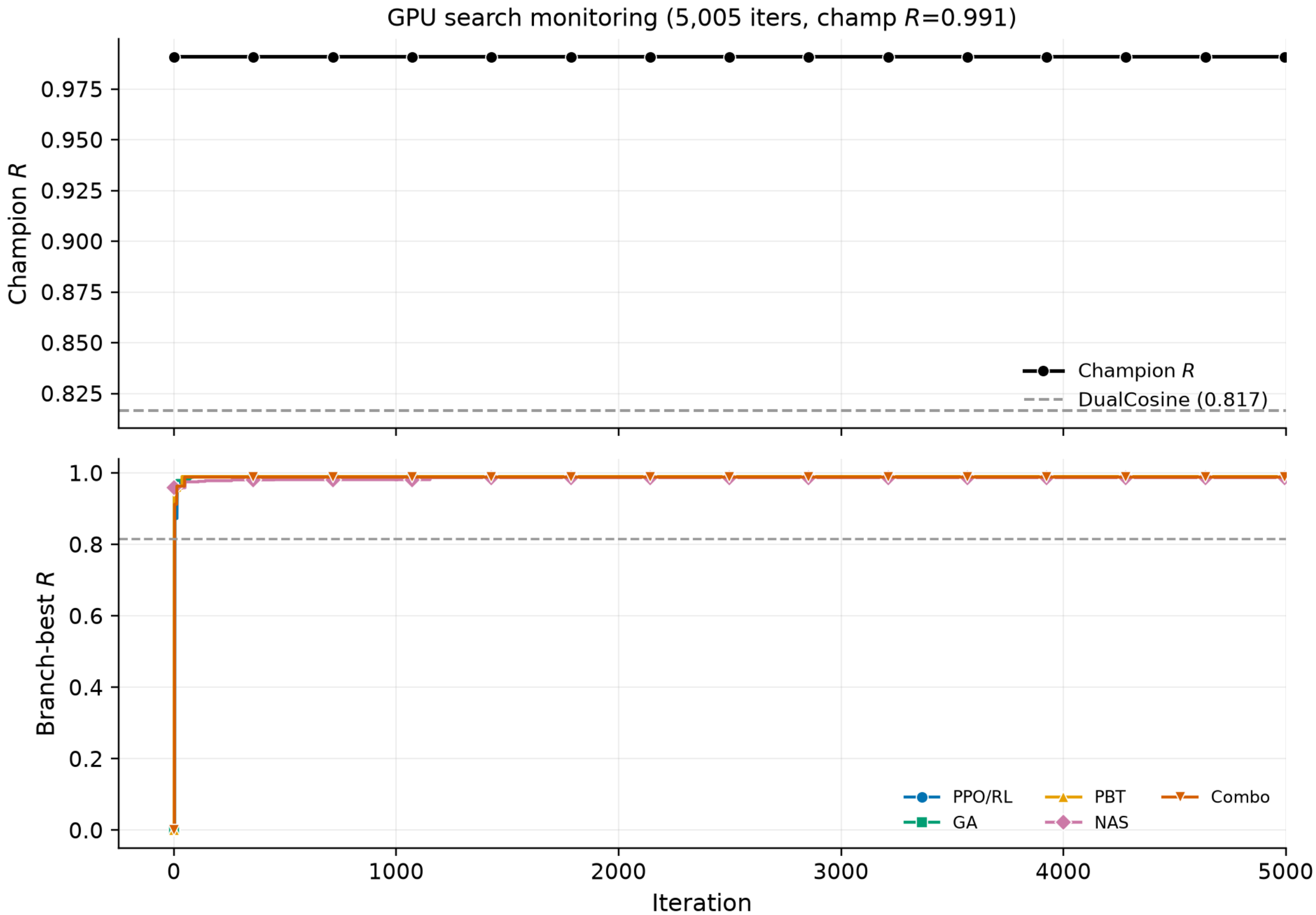

Figure 15: Full-width search monitoring panel at the 5k-gate freeze (first 5,000 iterations): champion R (top) and branch-best trajectories (bottom). Markers and Okabe-Ito colors match Figure 14. Source: `history.jsonl`.

Figure 16: Per-trial residual scatter by search branch for the first 5,000 iterations at the 5k-gate freeze, with champion overlay. Branch markers match Figure 14.

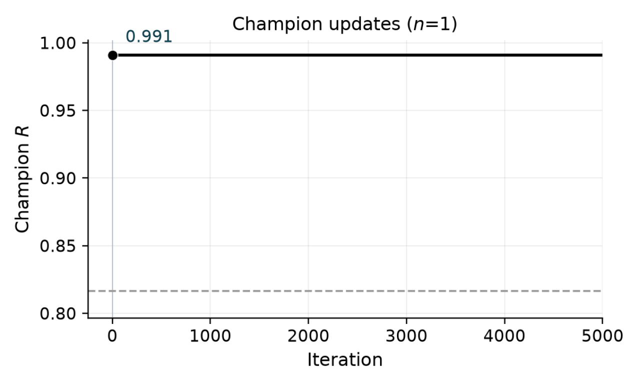

Figure 17: Champion update timeline for the first 5,000 iterations at the 5k-gate freeze. Sparse R labels mark first, mid-peak, and latest improvements for twocolumn legibility.

Harvest JSONs live in the companion meta repository. Source PDFs were downloaded for every bibliography entry (`artifacts/literature_oa/pdfs/`). Paywalled-only items were dropped or replaced.

Paper drafting. Section planning, subsection drafts, and revision passes used Klaut `research_paper_*` tools with local Ollama (`qwen3.5:9b`) when available, under an explicit scientific workflow (plan → write → revise → export) into an arXiv-style twocolumn template. AI drafts were treated as scaffolding. Claims, numbers, and citations were checked against harvest records, PDF analyses, and measured artifacts. Final prose was edited for DSP terminology and to remove generic filler.

Why AI was used. (1) Scale literature triage across DSP and AutoML without pretending every high-cite hit is in-domain. (2) Accelerate IMRaD scaffolding so human effort focuses on metric honesty (residual vs DualCosine, family hardness, gate vs final 5k-gate search claims). (3) Keep an auditable tool chain (plans, PDF snippet analyses, citation flags) over opaque chat-only writing.

Where AI was *not* used. Model fitting, residual scoring, DualCosine baselines, lit-combo / hybrid GPU search, and inference-time benchmarks are conventional code (Rust / PyTorch) with deterministic logs and JSON artifacts. AI did not invent experimental numbers.

Other tools. Rust/`cargo` for the procedural bench and meta binaries. PyTorch CUDA for hybrid GA–PPO search. matplotlib for figures. MiKTeX `pdflatex` for PDF builds. GitHub for versioned papers and artifacts.

8 Broader impact and reproducibility

Broader impact. This work targets cycle-local wavetable seam repair for virtual-analog / software-instrument workflows. Misuse potential is low. The method does not enhance speech for surveillance, does not remove watermarks from protected media, and operates only on short synthetic periods. Training and search use procedural sine+cliff generators. No private studio stems are required for the reported tables. The main resource cost is GPU time on a consumer RTX 3090 (order of hours for the 5k-gate freeze, Table 11). Authors should not over-claim speech enhancement or production mastering quality from these metrics.

Reproducibility. Holdout seed `20260719`. Hybrid search seed `1902771841`. Primary code: `https://github.com/reeldemo/reelsynth` (`scripts/overnight_gpu_rl_arch.py`, `scripts/bench_sota_matrix.py`, `scripts/metrics_snr_sdr.py`, `scripts/bench_va_seam_blep.py`). Paper companion: `https://github.com/reeldemo/denoise-opt-meta`. Dependencies for the CUDA benches: Python 3.12, PyTorch 2.6+CUDA 12.4, NumPy/matplotlib. Rust/`cargo` builds the separate `sound_bench` hardness probe. JSON artifacts under `brand/artifacts/` (`sota_matrix.json`, `canonical_eval_dataset/`, search `history.jsonl`, `va_seam_blep.json`) freeze the numbers in Section 5. PESQ/STOI are omitted for non-speech cycles (domain mismatch). MUSHRA was not run.

8.1 Funding

This research received no external funding.

8.2 Author contributions (CRediT)

Julian M. Kleber: Conceptualization, Methodology, Software, Investigation, Formal analysis, Data curation, Writing (original draft), Writing (review and editing), Visualization.

8.3 Data availability

Frozen evaluation artifacts (JSON matrices, search history, fitted champions) are published with the companion repositories: `https://github.com/reeldemo/reelsynth` under `brand/artifacts/`, and `https://github.com/reeldemo/denoise-opt-meta` under `artifacts/` and `paper/v7/figures/`. All reported tables regenerate from those files plus the scripts named above. No private studio recordings were used.

8.4 Competing interests

None declared.

9 Limitations

Scope and evaluation limits. The prolonged residual R is defined against a procedural no-cliff sibling and does not capture perceptual quality directly. Formal listening tests (MOS, MUSHRA) are outside the scope of this work and are deferred to future studies. PESQ and STOI are excluded because the primary cycles are non-speech synthetic wavetable tiles; speech-quality metrics on sine+cliff families would be a domain mismatch. MUSHRA was not run (requires human listeners). Any perceptual or speech-proxy secondary study would need separately labeled open-access speech snippets and remains out of scope for this narrow-seam paper.

- Residual uses a procedural ideal (no-cliff sibling). Perceptual MOS and listening panels are out of scope.
- Bake metrics leave formal listening tests and speech quality metrics to separate studies.
- **PESQ / STOI / MUSHRA deferred.** Primary cycles are non-speech wavetable tiles. Speech metrics would domain-mismatch sine+cliff families, so PESQ/STOI numbers are absent here (none invented on synthetic tiles). MUSHRA needs human listeners and was *not run*.
- **No LibriSpeech / MUSDB.** Wavetable-native realism uses true ReelSynth-exported factory periods (primary) and external AKWF CC0 single-cycle WAVs (secondary) under the wrap protocol. Procedural stand-ins are tertiary smoke only. Speech/music enhancement corpora remain off by design.
- Classical poly seam fitter is reported; lightweight SSM seq models are deferred (timebox); LSTM remains the supervised seq ceiling.
- On exported factory / Rust tiles the meta favorite can trail no-bake or FIR (domain shift). Transfer win-rates are frozen in `transfer_failures.json`; we do not claim universal factory superiority.

- The CUDA search runner scores Python `make_batch` sine+cliff draws. A separate Rust `sound_bench` export of 20 family tiles (byte-aligned generator) is a secondary transfer block. The Python generative SOTA matrix is the primary multi-family table.
- Per-trial architecture width is capped for search throughput. Full Demucs-/diffusion-scale stacks were screened out (domain mismatch vs L=256 periodic seams).
- We do not claim wrap-closure or hybrid-search convergence theorems. Empirical gains are protocol-bound. Endpoint-pin shows wrap-jump can be zeroed while R falls.
- The 5k-gate freeze reports measured champion and branch-best residuals. Isolated GA/PPO/GA+PPO/full re-runs use a short labeled budget (same search seed). Those short runs are a different protocol from the multi-hour freeze.
- Classical VA/BLEP lineage cites are OA author or proceedings PDFs (`stilson1996`, `nam2009polyblep`, `esqueda2016blamp`). Cycle-local BLIT/BLEP, PolyBLEP, and BLAMP seam repairs are now scored (`va_seam_blep.json`); they beat DualCosine on prolonged R in places but trail no-bake / FIR / the meta favorite on the value-cliff geometry.

9.1 Outlook: follow-up publication

A self-contained follow-up can treat *which sounds fail* as the primary object of study: stratified residual heatmaps over the ten procedural families, cliff-size conditioning, and family-specialist or mixture-of-experts denoisers selected by the same RL–GA outer loop. That paper would cite the present mean-R hybrid protocol as the baseline and ask whether beating $R > 0.999$ is mainly an architecture problem or a coverage problem over hard families.

10 Conclusion

DenoiseOpt casts wavetable wrap discontinuity as cycle-local seam repair under a prolonged residual R, with a hybrid GA–PPO(+PBT) outer loop over bake cells. At the 5k-gate search freeze the champion reaches $R \approx 0.9909$ versus DualCosine ≈ 0.8166. On the frozen holdout a compact favorite reaches $R \approx 0.991$ versus DualCosine $R \approx 0.825$; across twenty multi-family draws, mean $\Delta R \approx +0.162$ (Wilcoxon $p \approx 8.9 \times 10^{-5}$). Hard-cliff strata keep favorite $R \approx 0.9915$ while DualCosine falls to ≈ 0.657. Same-generator N2N / seq baselines bound the supervised ceiling. Wavetable-native transfer covers factory and OA instrument cycles under the wrap protocol. Scope remains wavetable seam restoration for DSP / synthesis venues. Code: https://github.com/reeldemo/reelsynth, https://github.com/reeldemo/denoise-opt-meta.

Acknowledgments

Compute for the CUDA hybrid search campaign used a local NVIDIA GeForce RTX 3090 workstation. Open literature was retrieved via the Klaut Research Gateway under open-access PDF constraints. No external collaborators contributed text, code, or experimental numbers for this preprint.

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